

Linear classification (logistic regression)

By

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Recap of last lecture

- We introduced the linear regression model, which is a model for solving the regression problem
- Now we will look at basic parametric modelling techniques, particularly
 - Linear regression (covered in last lecture)
 - Logistic regression
- Linear regression
 - A loss-based perspective, using least squares error
 - A closed form solution was derived
 - One-hot encoding to handle categorical inputs
- We will see that in logistic regression, we will not obtain a closed form solution

How to handle **categorical input** variables?

- We had mentioned earlier that input variables $\mathbf{x}$ can be numerical, categorical, or mixed
- Assume that an input variable is categorical and takes only **two classes**, say **A** and **B**

- We can represent such an input variable x using 1 and 0

$$x = \begin{cases} 0, & \text{if } \mathbf{A} \\ 1, & \text{if } \mathbf{B} \end{cases}$$

- For linear regression, the model effectively looks like

$$y = \theta_0 + \theta_1 x + \epsilon = \begin{cases} \theta_0 + \epsilon, & \text{if } \mathbf{A} \\ \theta_0 + \theta_1 + \epsilon, & \text{if } \mathbf{B} \end{cases}$$

- If the input is a categorical variable with **more than two classes**, let's say **A**, **B**, **C**, and **D**, use **one-hot encoding**

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \text{ if } \mathbf{A}, \quad \mathbf{x} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \text{ if } \mathbf{B}, \quad \mathbf{x} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \text{ if } \mathbf{C}, \quad \mathbf{x} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \text{ if } \mathbf{D}$$

A statistical view of the Classification problem

- Classification $\rightarrow$ learn relationships between some input variables $\mathbf{x} = [x_1 \ x_2 \ \dots \ x_p]^T$ and a **categorical** output y
- The goal in classification is to take an input vector $\mathbf{x}$ and to **assign it to one of M discrete classes $1, 2, \dots, M$**

- From a statistical perspective, classification amounts to predicting the **conditional class probabilities**

$$p(y = m | \mathbf{x}) \quad y \rightarrow 1, 2, \dots, M$$

- $p(y = m | \mathbf{x})$ describes **the probability for class m given that we know the input $\mathbf{x}$**

- A probability over output y implies **the output label y is a random variable (r.v.)**

- We consider y as a r.v. because the data (from real world) will always involve a certain amount of randomness (much like the output from linear regression that was probabilistic due to random error ϵ)



A statistical view of the Classification problem

- How to construct a classifier which can not only predict classes but also learn the class probabilities $p(y | \mathbf{x})$?
- Consider the simplest case of binary classification ($M = 2$) and $y = -1$ or 1
- In this **binary classification** case

$$p(y = 1|\mathbf{x}) \text{ will be modelled by } g(\mathbf{x})$$

- By the laws of probability,

$$p(y = 1|\mathbf{x}) + p(y = -1|\mathbf{x}) = 1$$

$$p(y = -1|\mathbf{x}) \text{ will be modelled by } 1 - g(\mathbf{x})$$

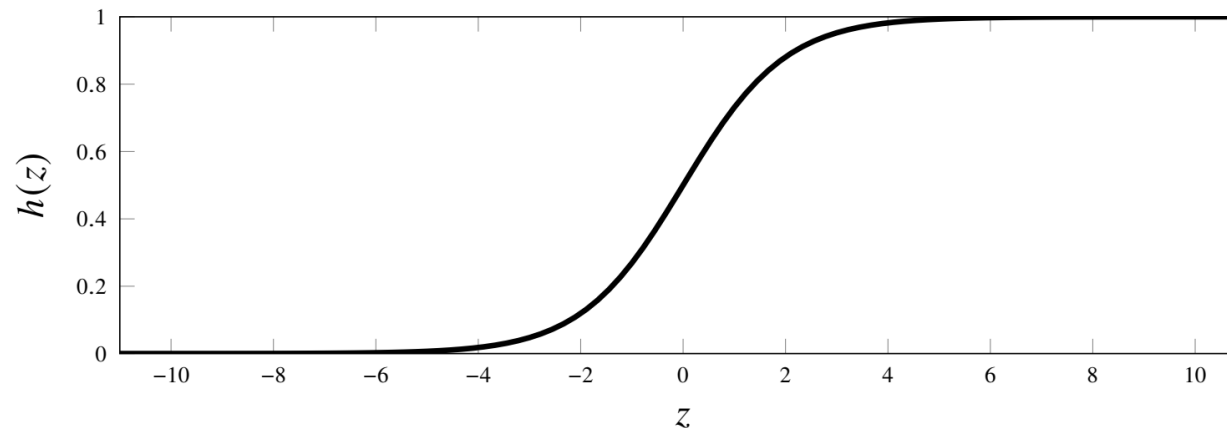
- Since $g(\mathbf{x})$ is a model for a probability, it is natural to require that $0 \leq g(\mathbf{x}) \leq 1$ for any $\mathbf{x}$
- For a multi-class problem, the classifier should return a vector-valued function $\mathbf{g}(\mathbf{x})$, where

$$\begin{bmatrix} p(y = 1|\mathbf{x}) \\ p(y = 2|\mathbf{x}) \\ \vdots \\ p(y = M|\mathbf{x}) \end{bmatrix} \text{ is modelled by } \begin{bmatrix} g_1(\mathbf{x}) \\ g_2(\mathbf{x}) \\ \vdots \\ g_M(\mathbf{x}) \end{bmatrix}$$

Since $\mathbf{g}(\mathbf{x})$ models a probability vector, each element $g_m(\mathbf{x}) \geq 0$ and $\sum_{m=1}^M g_m(\mathbf{x}) = 1$

Logistic Regression model for binary classification

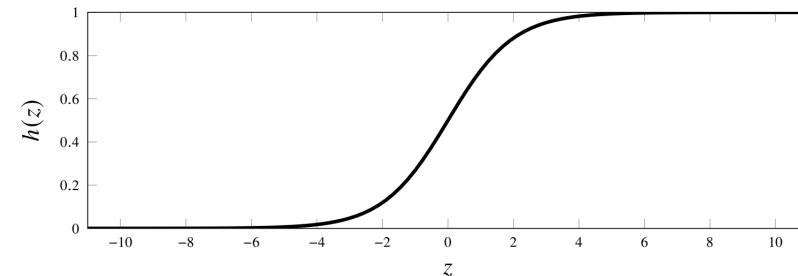
- Logistic regression can be viewed as an **extension of linear regression** that does (binary) **classification** (instead of regression)
- We wish to learn a function $g(\mathbf{x})$ that approximates the conditional probability of the **positive** class, $p(y = 1|\mathbf{x})$
- **Idea of Logistic Regression**: we start with the linear regression model which, without the noise term ϵ
 - Define **logit**, $z = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_p x_p = \mathbf{x}^T \boldsymbol{\theta}$
 - Logit takes values on the entire real line, but we need a function that returns a value in the interval $[0, 1]$
 - **Squash** the logit $z = \mathbf{x}^T \boldsymbol{\theta}$ into the interval $[0, 1]$ by using the **logistic function**, $h(z) = \frac{e^z}{1+e^z}$



Logistic Regression

- **Idea of Logistic Regression**: we start with the linear regression model which, without the noise term
 - Define **logit**, $z = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_p x_p = \mathbf{x}^T \boldsymbol{\theta}$
 - Logit takes values on the entire real line, but we need a function that returns a value in the interval $[0, 1]$
 - **Squash** the logit $z = \mathbf{x}^T \boldsymbol{\theta}$ into the interval $[0, 1]$ by using the **logistic function** $h(z) = \frac{e^z}{1+e^z}$
- Recall that $g(\mathbf{x})$ was used to model for $p(y = 1|\mathbf{x})$
- Using the logistic function for $g(\mathbf{x})$ restricts the values between 0 and 1 and can be interpreted as a probability

$$g(\mathbf{x}; \boldsymbol{\theta}) = \frac{e^{\mathbf{x}^T \boldsymbol{\theta}}}{1 + e^{\mathbf{x}^T \boldsymbol{\theta}}}$$



- It implicitly means that a model for $p(y = -1|\mathbf{x})$ is

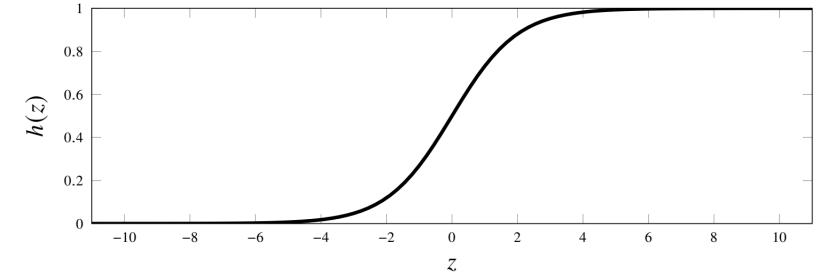
$$1 - g(\mathbf{x}; \boldsymbol{\theta}) = 1 - \frac{e^{\mathbf{x}^T \boldsymbol{\theta}}}{1 + e^{\mathbf{x}^T \boldsymbol{\theta}}} = \frac{1}{1 + e^{\mathbf{x}^T \boldsymbol{\theta}}} = \frac{e^{-\mathbf{x}^T \boldsymbol{\theta}}}{1 + e^{-\mathbf{x}^T \boldsymbol{\theta}}}$$

Logistic Regression

- **Logistic Regression**: Essentially linear regression appended with logistic function
 - **Logit**, $z = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_p x_p = \mathbf{x}^T \boldsymbol{\theta}$
 - $p(y = 1|\mathbf{x}; \boldsymbol{\theta}) = g(\mathbf{x}; \boldsymbol{\theta}) = \frac{e^{\mathbf{x}^T \boldsymbol{\theta}}}{1 + e^{\mathbf{x}^T \boldsymbol{\theta}}}$, $p(y = -1|\mathbf{x}; \boldsymbol{\theta}) = 1 - g(\mathbf{x}; \boldsymbol{\theta}) = \frac{e^{-\mathbf{x}^T \boldsymbol{\theta}}}{1 + e^{-\mathbf{x}^T \boldsymbol{\theta}}}$
- Logistic regression is a method for classification, not regression!
- The randomness in classification is statistically modelled by the class probability $p(y = m|\mathbf{x})$, instead of additive noise ϵ
- Like linear regression, logistic regression is also a parametric model, and we learn the parameters $\boldsymbol{\theta}$ from training data

Training binary classification model with Maximum Likelihood

- Logistic function is a **nonlinear** function
- Therefore, a closed-form solution to logistic regression cannot be derived



- Maximum likelihood perspective of learning θ from training data

$$\hat{\theta} = \operatorname{argmax}_{\theta} p(\mathbf{y}|\mathbf{X}; \theta)$$

- Similar to linear regression, we assume that the **training data points are independent**, and we consider the **logarithm of the likelihood function** for numerical reasons

$$\hat{\theta} = \operatorname{argmax}_{\theta} \ln p(\mathbf{y}|\mathbf{X}; \theta) = \operatorname{argmax}_{\theta} \sum_{i=1}^N \ln \left(p(y^{(i)}|\mathbf{x}^{(i)}; \theta) \right) = \operatorname{argmin}_{\theta} \sum_{i=1}^N -\ln \left(p(y^{(i)}|\mathbf{x}^{(i)}; \theta) \right)$$

- Note that $p(y = 1|\mathbf{x}; \theta)$ is modelled using $g(\mathbf{x}; \theta)$ which implies

$$-\ln p(y^{(i)}|\mathbf{x}^{(i)}; \theta) = \begin{cases} -\ln g(\mathbf{x}^{(i)}; \theta) & \text{if } y^{(i)} = 1 \\ -\ln (1 - g(\mathbf{x}^{(i)}; \theta)) & \text{if } y^{(i)} = -1 \end{cases}$$

Training binary classification model with Maximum Likelihood

- Assume that the **training data points are independent**, and we consider the **logarithm of the likelihood function** for numerical reasons

$$\hat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} \ln p(\mathbf{y}|\mathbf{X}; \boldsymbol{\theta}) = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} \sum_{i=1}^N \ln \left(p(y^{(i)}|\mathbf{x}^{(i)}; \boldsymbol{\theta}) \right) = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} \sum_{i=1}^N -\ln \left(p(y^{(i)}|\mathbf{x}^{(i)}; \boldsymbol{\theta}) \right)$$

- $p(y = 1|\mathbf{x}; \boldsymbol{\theta})$ is modelled using $g(\mathbf{x}; \boldsymbol{\theta})$

$$-\ln p(y^{(i)}|\mathbf{x}^{(i)}; \boldsymbol{\theta}) = \begin{cases} -\ln g(\mathbf{x}^{(i)}; \boldsymbol{\theta}) & \text{if } y^{(i)} = 1 \\ -\ln (1 - g(\mathbf{x}^{(i)}; \boldsymbol{\theta})) & \text{if } y^{(i)} = -1 \end{cases}$$

Cross-entropy loss function, $L(y^{(i)}, g(\mathbf{x}^{(i)}; \boldsymbol{\theta}))$

- Cross entropy loss can be used for **any binary classifier**, not just logistic regression, that predicts class probabilities $g(\mathbf{x}; \boldsymbol{\theta})$
- The corresponding **cost function** (or average loss function)

$$J(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^N \begin{cases} -\ln g(\mathbf{x}^{(i)}; \boldsymbol{\theta}) & \text{if } y^{(i)} = 1 \\ -\ln (1 - g(\mathbf{x}^{(i)}; \boldsymbol{\theta})) & \text{if } y^{(i)} = -1 \end{cases}$$

Training **Logistic Regression** model with Maximum Likelihood

- We can write the cost function in more detail for logistic regression

$$\text{For } y^{(i)} = 1, \quad g(\mathbf{x}^{(i)}; \boldsymbol{\theta}) = \frac{e^{(\mathbf{x}^{(i)})^T \boldsymbol{\theta}}}{1 + e^{(\mathbf{x}^{(i)})^T \boldsymbol{\theta}}} = \frac{e^{y^{(i)}(\mathbf{x}^{(i)})^T \boldsymbol{\theta}}}{1 + e^{y^{(i)}(\mathbf{x}^{(i)})^T \boldsymbol{\theta}}}$$

$$\text{For } y^{(i)} = -1, \quad 1 - g(\mathbf{x}^{(i)}; \boldsymbol{\theta}) = \frac{1}{1 + e^{(\mathbf{x}^{(i)})^T \boldsymbol{\theta}}} = \frac{e^{-(\mathbf{x}^{(i)})^T \boldsymbol{\theta}}}{1 + e^{-(\mathbf{x}^{(i)})^T \boldsymbol{\theta}}} = \frac{e^{y^{(i)}(\mathbf{x}^{(i)})^T \boldsymbol{\theta}}}{1 + e^{y^{(i)}(\mathbf{x}^{(i)})^T \boldsymbol{\theta}}}$$

- Hence, we get the **same expression in both cases** and can write the cost function compactly as:

$$\begin{aligned} J(\boldsymbol{\theta}) &= \frac{1}{N} \sum_{i=1}^N \begin{cases} -\ln g(\mathbf{x}^{(i)}; \boldsymbol{\theta}) & \text{if } y^{(i)} = 1 \\ -\ln (1 - g(\mathbf{x}^{(i)}; \boldsymbol{\theta})) & \text{if } y^{(i)} = -1 \end{cases} \\ &= \frac{1}{N} \sum_{i=1}^N -\ln \frac{e^{y^{(i)}(\mathbf{x}^{(i)})^T \boldsymbol{\theta}}}{1 + e^{y^{(i)}(\mathbf{x}^{(i)})^T \boldsymbol{\theta}}} = \frac{1}{N} \sum_{i=1}^N -\ln \frac{1}{1 + e^{-y^{(i)}(\mathbf{x}^{(i)})^T \boldsymbol{\theta}}} = \frac{1}{N} \sum_{i=1}^N \ln(1 + e^{-y^{(i)}(\mathbf{x}^{(i)})^T \boldsymbol{\theta}}) \end{aligned}$$

Training **Logistic Regression** model with Maximum Likelihood

- Cost function in logistic regression is given by:

$$J(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^N \underbrace{\ln(1 + e^{-y^{(i)}(\mathbf{x}^{(i)})^T \boldsymbol{\theta}})}_{\text{Logistic loss function, } L(y^{(i)}, \mathbf{x}^{(i)}; \boldsymbol{\theta})}$$

- The **logistic loss** $L(y^{(i)}, \mathbf{x}^{(i)}; \boldsymbol{\theta})$ above is a special case of the cross-entropy loss
- Learning a logistic regression model thus amounts to solving the optimization problem:

$$\hat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} J(\boldsymbol{\theta}) = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} \frac{1}{N} \sum_{i=1}^N \ln(1 + e^{-y^{(i)}(\mathbf{x}^{(i)})^T \boldsymbol{\theta}})$$

- Contrary to linear regression with squared error loss, the above problem **has no closed-form solution**, so we have to use **numerical optimization** instead

Predictions using Logistic Regression

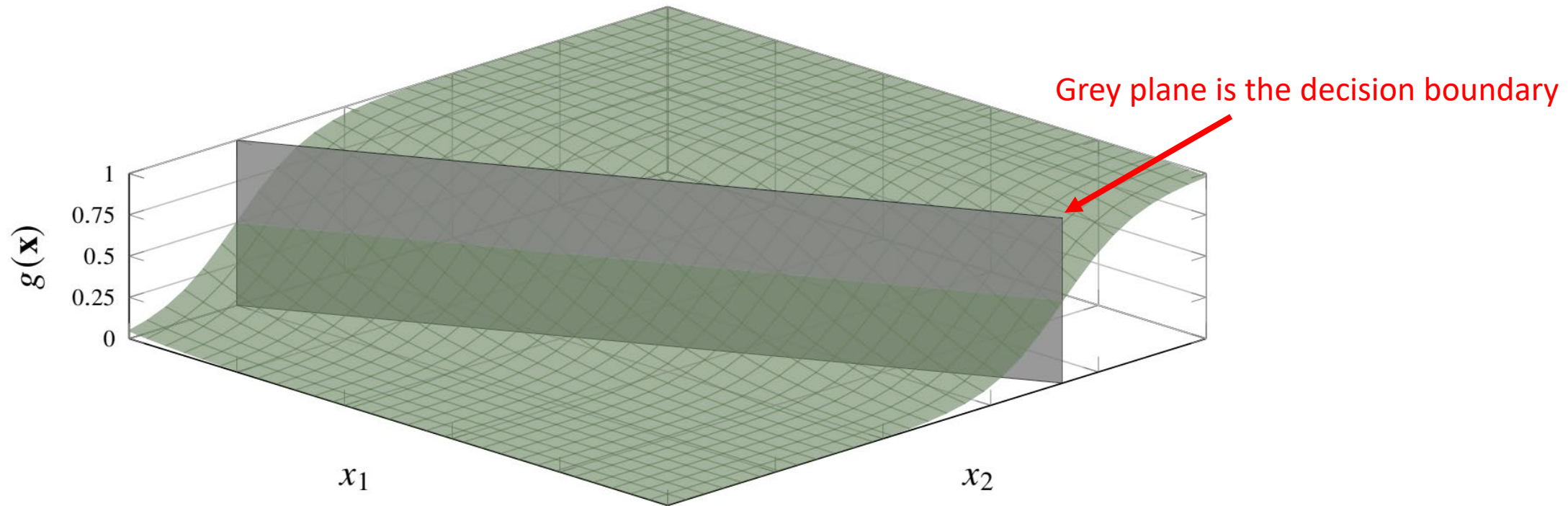
- Logistic regression predicts class probabilities for a test input $\mathbf{x}^*$
 - by first learning θ from training data, and
 - then computing $g(\mathbf{x}^*)$, which is the model for $p(y^* = 1|\mathbf{x}^*)$
- However, sometimes we want to make a “hard” prediction for the test input $\mathbf{x}^*$
 - E.g., whether is $\hat{y}(\mathbf{x}^*) = 1$ or $\hat{y}(\mathbf{x}^*) = -1$ in binary classification?
 - Recall, in k NN and decision trees, we made “hard” predictions
- To make hard predictions with logistic regression model, we add a final step, in which the predicted probabilities are turned into a class prediction
- The most common approach is to let $\hat{y}(\mathbf{x}^*)$ be the *most probable class* ← the class having the highest probability
- For binary classification, we can express this as:

$r = 0.5$ minimises the so-called *misclassification rate*

$$\hat{y}(\mathbf{x}^*) = \begin{cases} 1 & \text{if } g(\mathbf{x}^*) > r \\ -1 & \text{if } g(\mathbf{x}^*) \leq r \end{cases} \text{ with decision threshold } r = 0.5 \text{ (why?)}$$

Decision Boundaries of Logistic Regression

- Decision boundary ← The point(s) where the prediction changes from from one class to another



- The decision boundary for binary classification can be computed by solving the equation $g(\mathbf{x}) = 1 - g(\mathbf{x})$ meaning $p(y = 1|\mathbf{x}; \boldsymbol{\theta}) = p(y = -1|\mathbf{x}; \boldsymbol{\theta})$
- The solutions to this equation are points in the input space for which the two classes are predicted to be equally probable

Decision Boundaries of Logistic Regression

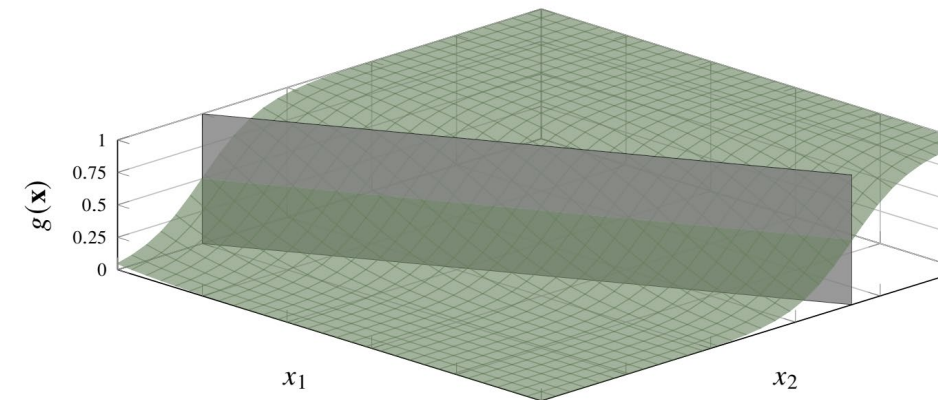
- The decision boundary for binary classification can be computed by solving the equation

$$g(\mathbf{x}) = 1 - g(\mathbf{x}) \quad \text{meaning} \quad p(y = 1|\mathbf{x}; \boldsymbol{\theta}) = p(y = -1|\mathbf{x}; \boldsymbol{\theta})$$

- The solutions to this equation are points in the input space for which the two classes are predicted to be equally probable
- For binary logistic regression, it means

$$\frac{e^{\mathbf{x}^T \boldsymbol{\theta}}}{1 + e^{\mathbf{x}^T \boldsymbol{\theta}}} = \frac{1}{1 + e^{\mathbf{x}^T \boldsymbol{\theta}}} \Leftrightarrow e^{\mathbf{x}^T \boldsymbol{\theta}} = 1 \Leftrightarrow \mathbf{x}^T \boldsymbol{\theta} = 0$$

- The equation $\mathbf{x}^T \boldsymbol{\theta} = 0$ parameterises a (linear) hyperplane
- Therefore, the decision boundaries in logistic regression always have the shape of a (linear) hyperplane



Prediction and Decision Boundaries of Logistic Regression

- For binary classification, we can express this as:

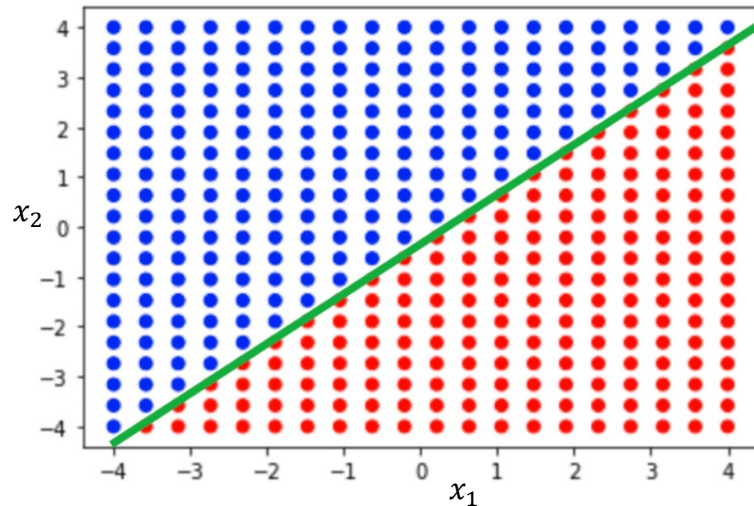
$$\hat{y}(\mathbf{x}^*) = \begin{cases} 1 & \text{if } g(\mathbf{x}^*) > r \\ -1 & \text{if } g(\mathbf{x}^*) \leq r \end{cases} \text{ with decision threshold } r = 0.5$$

- Choosing $r = 0.5$ minimises the so-called misclassification rate
- The decision boundary for logistic regression lies at $\mathbf{x}^T \boldsymbol{\theta} = 0$
 $\Rightarrow$ The **sign of the expression $\mathbf{x}^T \boldsymbol{\theta}$** determines if we are predicting the positive (1) or the negative (-1) class
- Compactly, one can write the test output prediction for a test input $\mathbf{x}^*$ from a logistic regression as

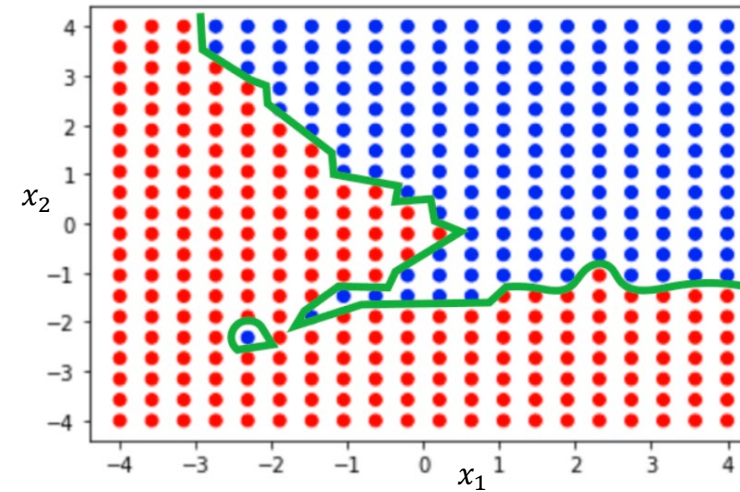
$$\hat{y}(\mathbf{x}^*) = \text{sign}(\mathbf{x}^{*T} \boldsymbol{\theta})$$

Linear vs Non-linear classifiers

- A classifier whose decision boundaries are linear hyperplanes is a *linear classifier*
- Logistic regression is a linear classifier
- k NN and Decision Trees are non-linear classifiers



Linear classifier



Non-Linear classifier

- Note that the term 'linear' has a different sense for linear regression and for linear classification
 - Linear regression is a model which is *linear in its parameters*,
 - Linear classifier is a model linear whose *decision boundaries are linear*

Logistic Regression for more than two classes

- For the binary problem, we used the logistic function to design a model for $g(\mathbf{x})$
 - $g(\mathbf{x})$ a scalar-valued function representing $p(y = 1 | \mathbf{x})$
- For a multi-class problem (M classes), the classifier should return a vector-valued function $\mathbf{g}(\mathbf{x})$, where

$$\begin{bmatrix} p(y = 1 | \mathbf{x}) \\ p(y = 2 | \mathbf{x}) \\ \vdots \\ p(y = M | \mathbf{x}) \end{bmatrix} \text{ is modelled by } \mathbf{g}(\mathbf{x}) = \begin{bmatrix} g_1(\mathbf{x}) \\ g_2(\mathbf{x}) \\ \vdots \\ g_M(\mathbf{x}) \end{bmatrix}$$

Since $\mathbf{g}(\mathbf{x})$ models a probability vector, each element $g_m(\mathbf{x}) \geq 0$ and $\sum_{m=1}^M g_m(\mathbf{x}) = 1$

- For this purpose, we define M different logits, $z_m = (\boldsymbol{\theta}^m)^T \mathbf{x}$, $m = 1, 2, \dots, M$
- The use the *softmax* function (a vector-valued generalization of logistic function)

$$\text{softmax}(\mathbf{z}) \triangleq \frac{1}{\sum_{m=1}^M e^{z_m}} \begin{bmatrix} e^{z_1} \\ e^{z_2} \\ \vdots \\ e^{z_M} \end{bmatrix}$$

- $\mathbf{z}$ is an M -dimensional vector
- $\text{softmax}(\mathbf{z})$ also returns a vector of the same dimension
- By construction, the output vector always sums to 1, and each element is always ≥ 0

Multi-class Logistic Regression model

- We have now combined **linear regression** and **softmax function** to model multi-class probabilities

$$\mathbf{g}(\mathbf{z}) = \text{softmax}(\mathbf{z}), \quad \text{where } \mathbf{z} = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_M \end{bmatrix} = \begin{bmatrix} (\boldsymbol{\theta}^1)^T \mathbf{x} \\ (\boldsymbol{\theta}^2)^T \mathbf{x} \\ \vdots \\ (\boldsymbol{\theta}^M)^T \mathbf{x} \end{bmatrix}$$

- Equivalently, we can write out the individual class probabilities, that is, the elements of the vector $g_m(\mathbf{x})$

$$g_m(\mathbf{x}) = \frac{e^{(\boldsymbol{\theta}^m)^T \mathbf{x}}}{\sum_{j=1}^M e^{(\boldsymbol{\theta}^j)^T \mathbf{x}}} \quad m = 1, 2, \dots, M$$

- This is the **multiclass** logistic regression model
- Note that this construction uses M parameter vectors $\boldsymbol{\theta}^1, \dots, \boldsymbol{\theta}^M$ (one for each class)
 - Note the number of parameters to learn grows with M